Network Science

Lecture 04: Communities and Graph Partitioning

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From Classifying Edges to Classifying Nodes and Groups

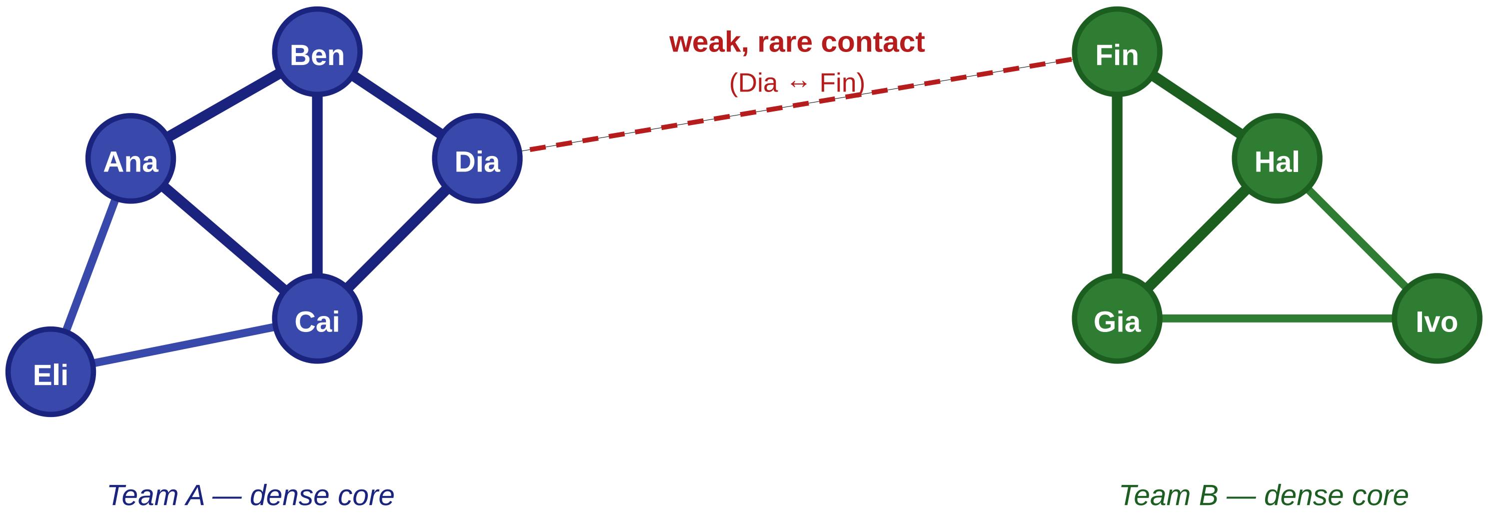
In Lecture 03 we asked which *edges* carried novel information — strong vs. weak, bridges vs. embedded. Today we ask which *nodes* occupy important positions, and how the graph splits into dense groups.

Let us return to the same workplace.

The Same Scenario, New Questions

A small workplace: five questions in one picture

nodes = employees · thick edges = close contacts · thin edges = acquaintances



Questions we'd like to answer from this picture

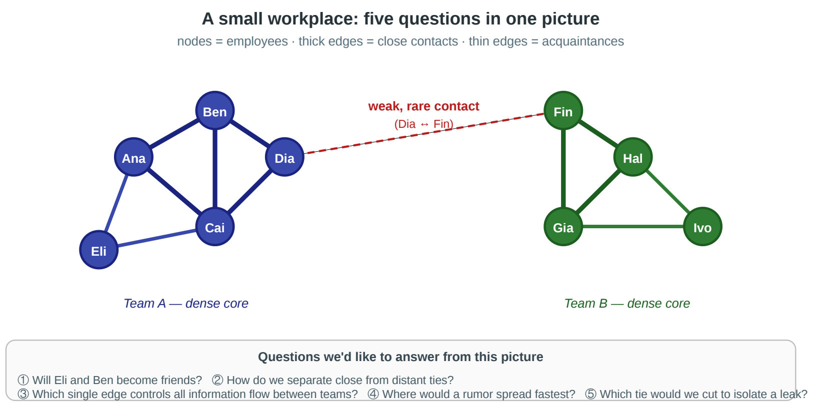
① Will Eli and Ben become friends? ② How do we separate close from distant ties?
③ Which single edge controls all information flow between teams? ④ Where would a rumor spread fastest? ⑤ Which tie would we cut to isolate a leak?

Same graph as last time — two teams linked by a single weak Dia–Fin contact. Lecture 03 asked about the edges. Today the questions are about the nodes and the groups.

Four Group-Level Questions

1. **Importance.** Who is the single most important person in this office — and what does "important" even mean?
2. **Boundaries.** Where is the line between team A and team B, and could an algorithm discover it *without being told the labels*?
3. **Uniqueness.** If we ran community detection on a 10 000-person company, would the answer be unique — or would different methods give different maps?
4. **Validation.** How do we know when a detected community is real rather than an artifact of the algorithm?

Every concept in today's lecture is a formal answer to one of these questions.



Theory vs. Measurement — Continued

The gap we opened in Lecture 03 does not close in Lecture 04. Every concept today has a theoretical form we cannot compute on real data and a measurable proxy we can.

Question	Theoretical construct	Measurable proxy
Who is most important?	*(no single answer — multiple theories)*	Degree / closeness / betweenness / eigenvector
What makes a group a group?	Community with high internal density	Modularity Q
Best partition?	Global modularity maximum (NP-hard)	Girvan–Newman / Louvain heuristics
Is a partition meaningful?	Agreement with ground-truth labels	Benchmark datasets (Zachary's karate club)

Takeaway

> L03 classified edges; L04 classifies nodes and groups. The modeling–measurement loop is the same: a theoretically clean objective is NP-hard, polynomial-time heuristics approximate it, and empirical benchmarks test whether the heuristics recover meaningful structure.

Roles and Social Capital

> **Guiding Question:** What kinds of structurally different positions can people occupy inside the same network?

Three Structural Positions

Embeddedness. A node whose neighbors are themselves densely interconnected. High clustering coefficient.

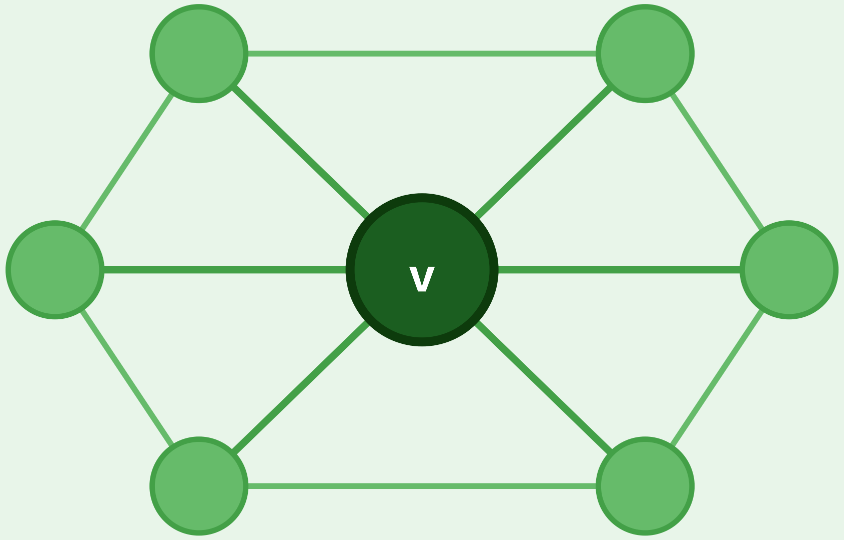
Brokerage. A node connecting otherwise separated groups. Low clustering coefficient, high betweenness.

Structural hole. The empty region in the network that a brokerage position spans.

Embedded vs. Broker

Embedded Node

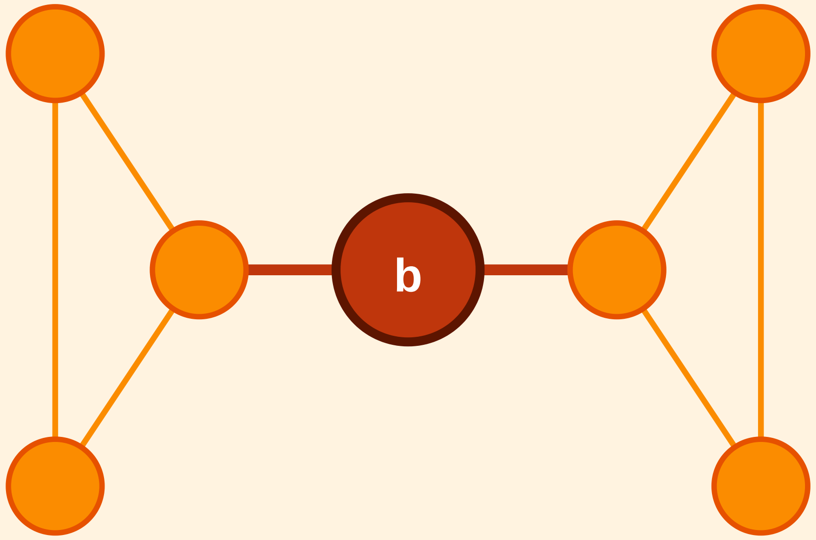
Dense, mutually connected neighbourhood



High clustering · trust · redundant info

Broker (Structural Hole)

Sole link between two separated clusters



Low clustering · high betweenness · novel info

The embedded node v sits inside a tightly connected neighborhood — its removal barely affects connectivity. The broker b is the only short link between two clusters — its removal would isolate them.

Takeaway

> "Importance" is not one thing. Brokerage, embeddedness, and degree capture different social advantages — and the same node can sit in multiple roles depending on the question.

Quick Check

Look again at the workplace figure (Ana, Ben, Cai, Dia, Eli on the left; Fin, Gia, Hal, Ivo on the right).

1. Which of these nine people is the clearest *broker*, and what structural feature makes them one?
2. Which is the most clearly *embedded*?
3. What would change structurally if Dia left the company tomorrow?

Quick Check — Solution

1. Broker: Dia — she is the only endpoint of the cross-team edge. Every shortest path between the two teams passes through her.

2. Embedded: Ana, Ben, and Cai are all embedded (they sit inside the same closed triangle). Ben is perhaps the most embedded, with all three neighbors (Ana, Cai, Dia) mutually connected.

3. If Dia leaves: the Dia–Fin edge disappears with her, and the two teams become *disconnected components*. The structural hole becomes absolute — no shortest path exists between Ana and Hal at all.

Centrality

> **Guiding Question:** How should we measure which nodes are important?

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$$\text{importance} \neq \text{degree alone}$$

Degree and Closeness Centrality

Degree centrality

$$C_D(v) = \frac{\deg(v)}{n-1}$$

Measures **local exposure** — how many direct contacts a node has. Fast to compute; blind to network position.

Example: The most-followed account on a social platform ranks high on degree — but "famous" and "structurally central" are different things.

Closeness centrality

$$C_C(v) = \frac{n-1}{\sum_{u \neq v} d(v,u)}$$

Measures **reach** — high when average shortest-path distance to all other nodes is small. Captures how fast information spreads from this node.

Example: An emergency-dispatch hub close to all districts has high closeness — any location can be reached in minimum average time.

Betweenness and Eigenvector Centrality

Betweenness centrality

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Measures **control** — fraction of all shortest paths that pass through v. σ_{st}: number of shortest s–t paths; $\sigma_{st}(v)$: those passing through v.

Example: A router at a continental internet exchange has high betweenness — all intercontinental traffic passes through it.

Eigenvector centrality

$$C_E(v) = \frac{1}{\lambda} \sum_{u} A_{vu}, C_E(u)$$

Measures **prestige** — recursive, a node is important if connected to other important nodes. A: adjacency matrix; λ: its leading eigenvalue.

PageRank (Google, 1998) is the directed, damped variant: prestige flows along directed citation/link edges, with a damping factor to handle dangling nodes.

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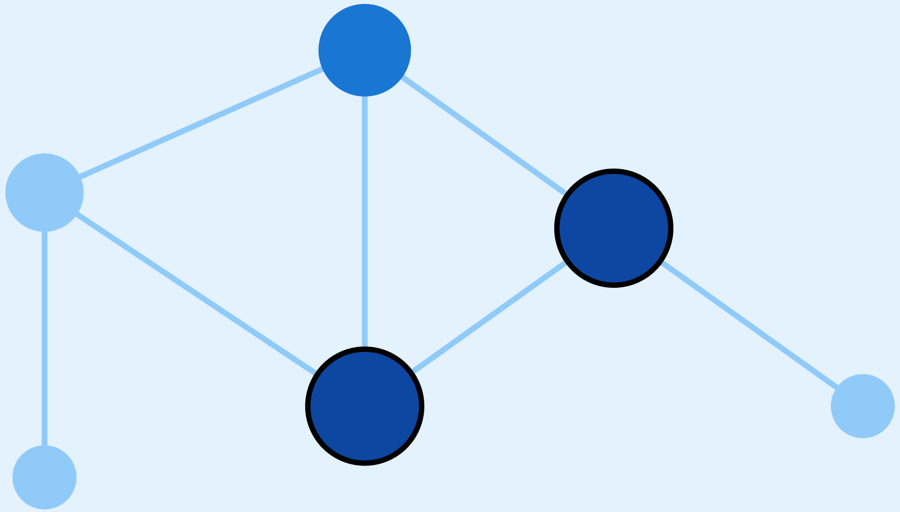
Same Graph, Four Views of Importance

Same graph, four notions of "important"

Degree

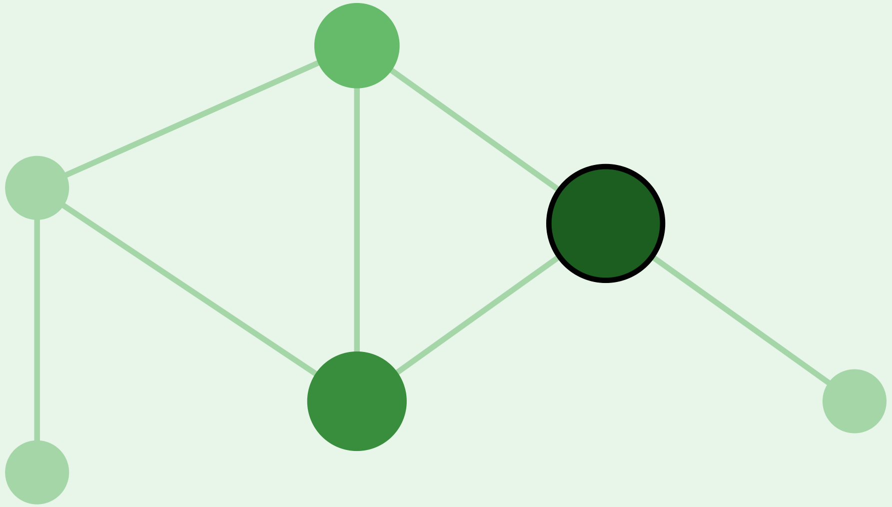
how many direct connections



node size ∝ degree

Closeness

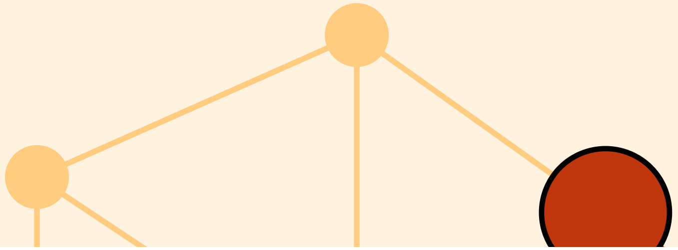
low average distance to all others



closer to center → larger

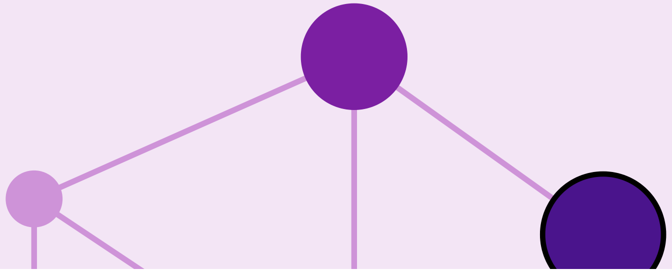
Betweenness

how often lies on shortest paths



Eigenvector

connected to other important nodes



Back to the Workplace

In the workplace figure, each measure picks a different winner:

- **Degree** → Ana (or Ben): most direct contacts inside team A's core
- **Closeness** → Dia: smallest average distance to everyone, because she sits on the only bridge
- **Betweenness** → Dia *and* Fin: every shortest path between the teams passes through both
- **Eigenvector** → Ana or Ben: connected to densely-connected others, which amplifies recursively

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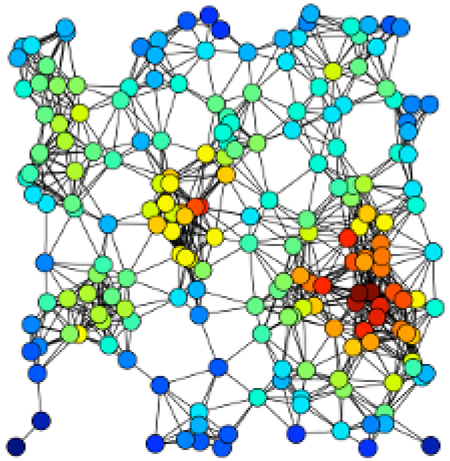
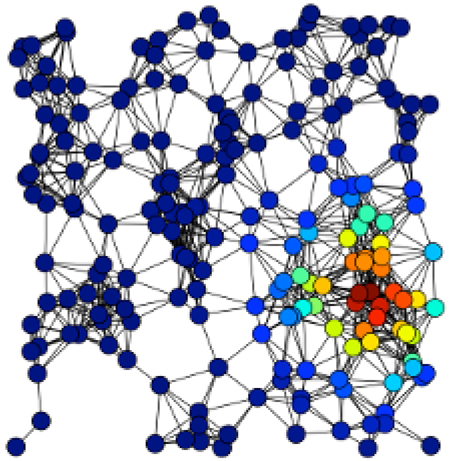
Real-World Examples



Source: Easley & Kleinberg 2010



Source: Easley & Kleinberg 2010

Left: Degree highlights raw connection counts. **Right:** Eigenvector centrality refines this — a node is important if it links to other important nodes. Google's PageRank is the directed variant of eigenvector centrality applied to the Web.

Takeaway

Centrality is always tied to a theory of influence, access, or control. Choosing a measure is choosing a theory — not collecting an objective fact.

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Quick Check

For each scenario, decide which centrality measure is most appropriate and justify briefly.

1. You want to identify the most influential gossip in a small village.
2. You want to find emergency-response hubs that can reach all districts within minimum time.
3. You want to detect which router, if attacked, would slow internet traffic between continents most.
4. You want to find the most prestigious researcher in a citation network.

Quick Check — Solution

1. **Degree centrality** — gossip is local, direct contacts matter.
2. **Closeness centrality** — minimizing average distance to all districts.
3. **Betweenness centrality** — the router's role is to lie on paths between continents.
4. **Eigenvector centrality / PageRank** — prestige flows from being cited by others who are themselves prestigious.

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What Counts as a Community?

> **Guiding Question:** Can we give "community" a mathematical definition precise enough that an algorithm could find one?

Working Definition

> **Community (informal).** A subset of nodes $S \subseteq V$ with
>
> - relatively many edges *inside* S
> - relatively few edges *leaving* S
> - density meaningfully higher than the surrounding graph.

Different formal definitions instantiate this intuition differently. The most influential is **modularity**, which compares observed internal density against a random null model.

Modularity — The Canonical Objective

Modularity Q of a partition $C_1, \ldots, C_k$ of a graph with m edges and adjacency matrix A:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

where k_i is the degree of node i and $\delta(c_i, c_j) = 1$ if i and j are in the same community, else 0.

Interpretation. For every within-community pair, compare the *observed* edge A_{ij} to the *expected* edge density $k_i k_j / 2m$ under a random rewiring that preserves degrees (the **configuration model**). Q is the total surplus of within-community edges over chance.

Q ranges roughly in $[-\frac{1}{2}, 1]$. Values in $[0.3, 0.7]$ typically indicate significant community structure; $Q \approx 0$ means no more internal density than expected by chance.

The Catch — Modularity Maximization Is NP-Hard

> **Result (Brandes et al., 2008).** Finding the partition that maximizes Q on a general graph is **NP-hard**. No polynomial-time algorithm for the exact optimum is known.

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Sound familiar? This is the exact same situation as **MaxSTC** in Lecture 03: a theoretically clean objective that is computationally out of reach on real data.

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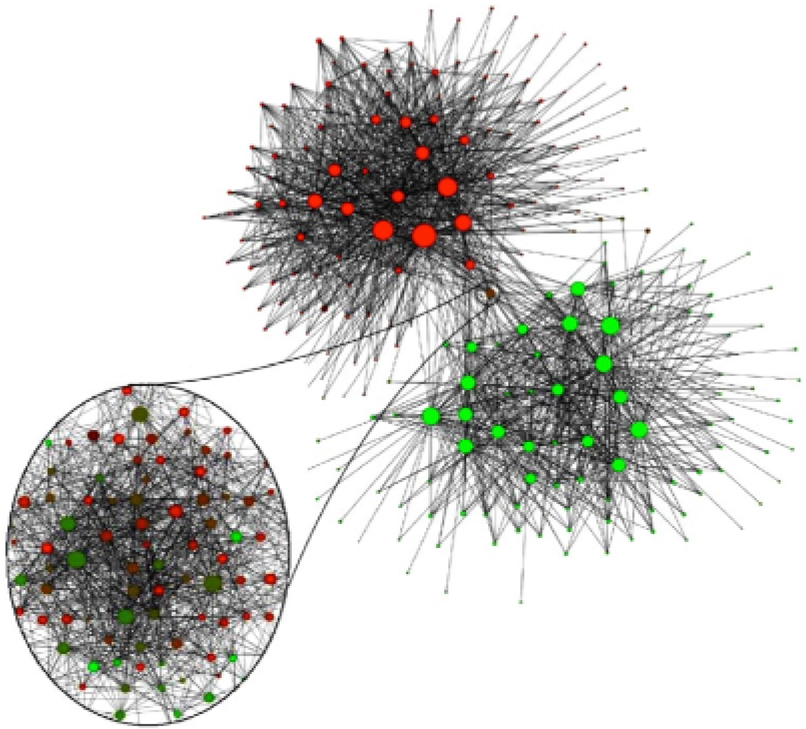
Consequence. The rest of the module introduces two families of polynomial-time *heuristics* that find partitions with high Q without guaranteeing the global maximum:

- **divisive** (Girvan–Newman): peel apart high-betweenness edges
- **agglomerative** (Louvain, Leiden): greedily merge nodes that increase Q

Example Cases



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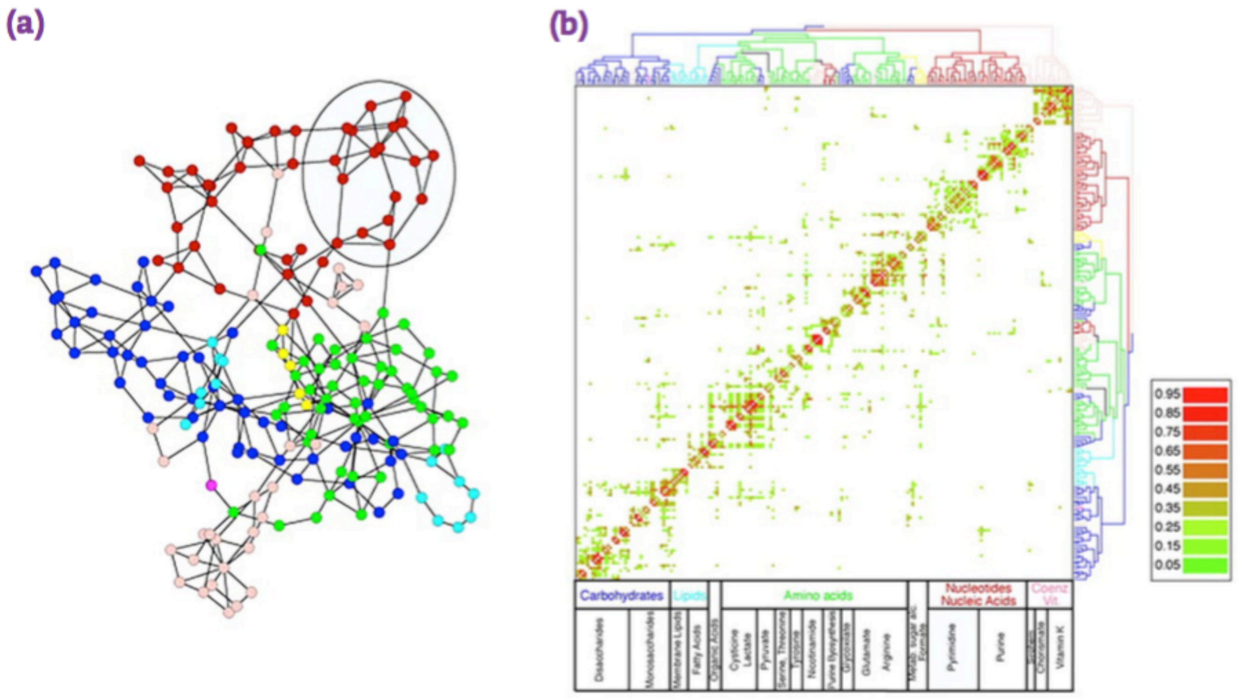


Source: Easley & Kleinberg 2010

Three different domains, the same structural pattern: dense internal connectivity and sparse cross-group connectivity. The Belgium phone graph reveals the linguistic divide between Flemish and Walloon speakers — an external label the algorithm was never told. Metabolic communities reflect functional pathway groupings.

Takeaway

> Modularity Q is the canonical mathematical definition of "community": excess internal density over a random-graph null model. Maximizing it exactly is NP-hard — so every practical community detection method is a heuristic.

Quick Check

A graph has two dense clusters of 50 nodes each connected by a single edge, plus one small triangle attached to one cluster via a single edge.

1. Roughly how many natural communities does this graph have?
2. Will modularity maximization find all of them?
3. If you asked a balanced 2-partition algorithm to split it into two equal halves, what might go wrong?

Quick Check — Solution

1. Three natural communities — the two dense clusters and the small triangle.

2. Modularity will find them on this small graph, but in large graphs modularity has a **resolution limit**: it systematically fails to detect communities smaller than roughly $\sqrt{2m}$ edges. Scale the two clusters up to 5000 nodes each and the triangle vanishes from view.

3. Balanced 2-partition is forced to produce equal-sized groups. It will either cut through one of the dense clusters or merge the triangle with the wrong cluster. The constraint distorts the answer.

Finding Communities — Divisive and Agglomerative

Guiding Question: If modularity maximization is NP-hard, how do we find good partitions in practice?

Two Strategies

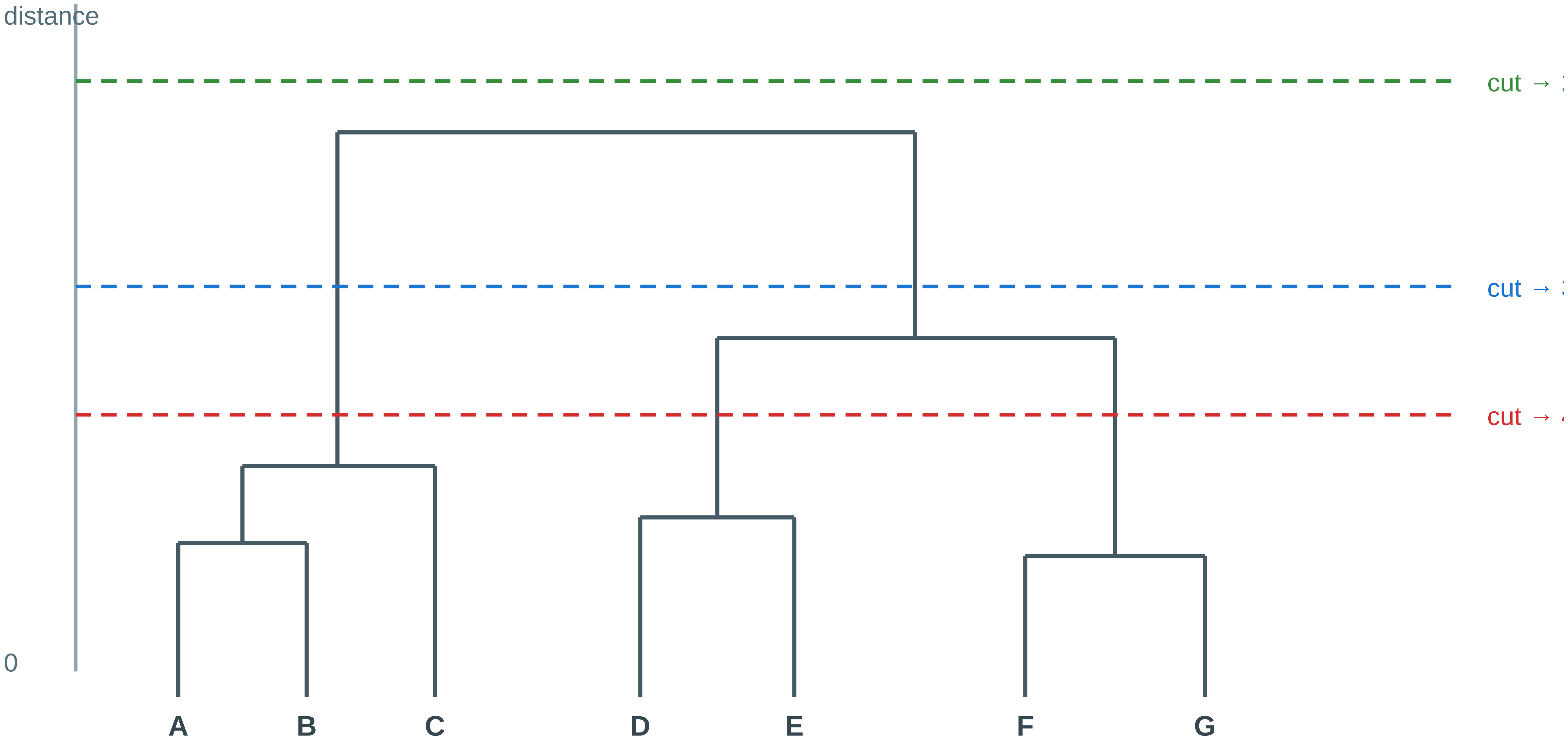
Divisive (top-down). Start with the whole graph as one cluster. Repeatedly remove edges that lie *between* communities, peeling the graph apart. Example: Girvan–Newman.

Agglomerative (bottom-up). Start with each node as its own cluster. Repeatedly merge the pair whose merge most *increases* modularity. Example: Louvain, Leiden.

Both strategies produce a hierarchy — a **dendrogram** — rather than a single flat partition. Where to cut the dendrogram is the analyst's modeling decision.

The Dendrogram as Output

Hierarchical clustering: the cutoff is a modeling choice



Each horizontal cut yields a different clustering — the analyst chooses the level

Girvan–Newman — The Divisive Algorithm

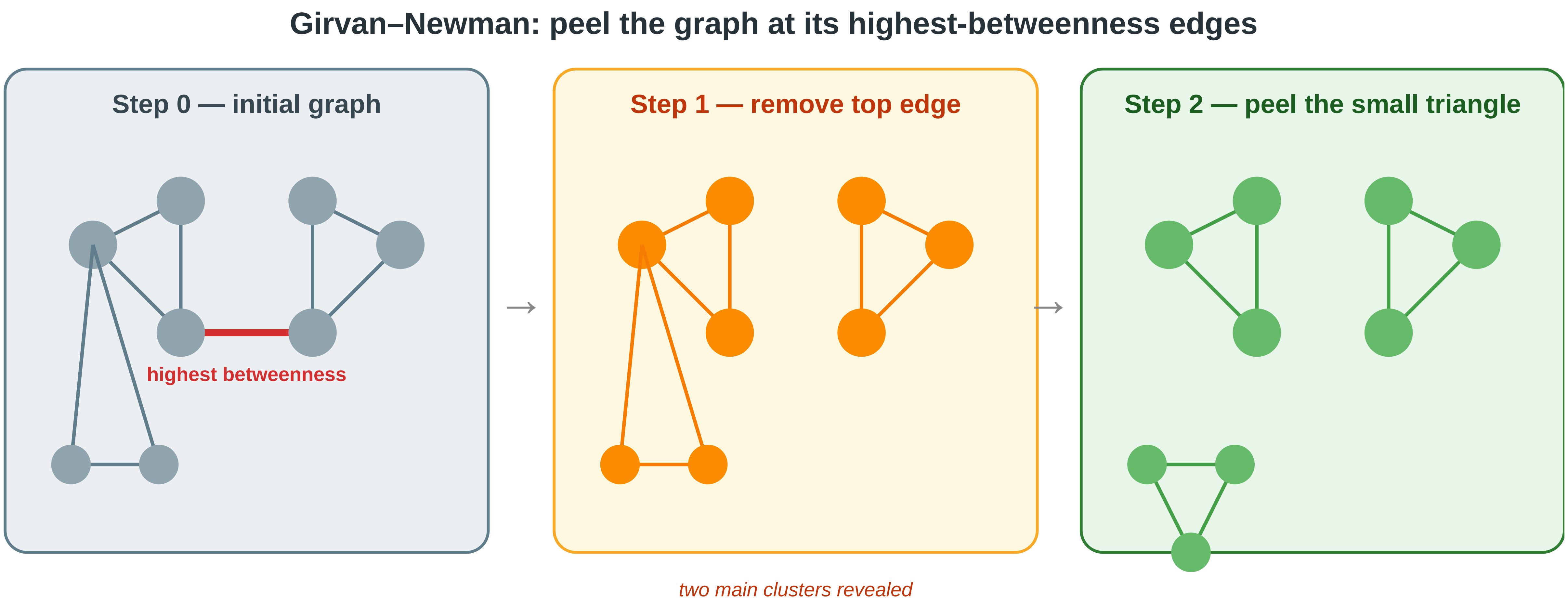
Algorithm: Girvan–Newman

(Input: Graph $G = (V, E)$)

1. Compute the **edge betweenness** of every edge — the fraction of all shortest paths that pass through it.
2. Remove the edge with the **highest** edge betweenness.
3. Recompute edge betweenness on the remaining graph.
4. Repeat until no edges remain — record the partition at each step.
5. Choose the partition that maximizes **modularity** Q.

Edges with high edge-betweenness are exactly the **local bridges** and **weak ties** from Lecture 03 — they carry many shortest paths because all cross-community communication must cross them. Girvan–Newman is an algorithmic version of Granovetter's weak-tie argument.

Worked Example



Step 0: the bridge edge has the highest betweenness because every path between the two clusters must cross it. **Step 1:** removing it reveals two main clusters but the small triangle remains weakly attached. **Step 2:** the next-highest-betweenness edge is the link to the triangle — removing it isolates the third community. Final partition: three communities.

Louvain and Leiden — Modern Practice

Louvain (Blondel et al., 2008). Agglomerative modularity maximizer.

Phase 1: For each node, try moving it to each neighbor's community; keep the move that most increases Q. Repeat until no move improves Q.

Phase 2: Contract each community into a super-node and repeat Phase 1 on the new graph. Iterate until Q converges.

Leiden (Traag et al., 2019). Fixes a subtle Louvain flaw (disconnected communities) and guarantees well-connected output at each level.

Why these dominate practice. Near-linear time in $|E|$; handle millions of nodes; directly optimize Q; no edge-betweenness recomputation.

Graph Partitioning — Cut-Based Methods

Min-cut / Max-flow (Ford–Fulkerson) Find the smallest set of edges whose removal disconnects two specified groups. Exact in polynomial time — but minimizes raw cut size, not normalized density.

Kernighan–Lin (1970) Local search: initialize a balanced 2-partition at random; iteratively swap node pairs to decrease the cut size. Fast in practice; converges to a local optimum.

Spectral partitioning Compute the Laplacian $L = D - A$ of the graph. The **Fiedler vector** (eigenvector of the second-smallest eigenvalue λ_2) encodes the natural 2-partition: nodes with positive entries go left, negative entries go right.

λ_2 is the **algebraic connectivity**: $\lambda_2 = 0$ means the graph is already disconnected. Spectral methods generalize to k communities by using the k smallest eigenvectors — the basis of **spectral clustering**.

Embedding-Based Community Detection

Node embeddings + clustering

> **node2vec (Grover & Leskovec, 2016)** Random-walk–based embedding: simulate biased random walks on the graph, then train Word2Vec–style skip-gram to produce a low-dimensional vector per node. Communities emerge as clusters in embedding space.

Pipeline: node2vec → k-means / DBSCAN → community labels

Hyperparameter p controls return probability (DFS-like, captures communities), q controls exploration (BFS-like, captures roles). Setting $p \gg q$ favors dense cluster detection.

Graph Neural Networks

> **GNN-based community detection** Train a GNN encoder (e.g. GCN, GraphSAGE) to produce embeddings that maximize an unsupervised objective — modularity, contrastive loss, or cluster assignment entropy. The decoder maps embeddings to community memberships.

Approach	Key idea
GRAPH-BERT	Self-supervised pre-training on subgraphs
DMGI	Contrastive multiplex embeddings
DMoN	Differentiable modularity as loss

GNNs naturally handle **overlapping communities** (soft cluster assignment) and **attributed graphs** (node features + topology jointly).

Algorithm Comparison

Method	Strategy	Complexity	Strength	Limitation
Min-cut	Cut minimization	$O(\|V\| \cdot \|E\|)$	Exact for 2-partition	Ignores density, favors small cut
Kernighan–Lin	Local swap	$O(\|V\|^2 \log \|V\|)$	Fast for balanced partition	Local optimum, fixed k
Spectral	Laplacian eigenvectors	$O(\|V\|^3)$ full / faster approx	Global structure, principled	Slow exact; needs k in advance
Girvan–Newman	Edge betweenness	$O(\|V\| \cdot \|E\|^2)$	Hierarchy, no k needed	Too slow for large graphs
Louvain / Leiden	Modularity greedy	$O(\|E\|)$ approx	Fast, scalable, no k needed	Resolution limit, local opt
node2vec + clustering	Random-walk embed	$O(\|E\|)$ walks + embed	Scales, uses node features	Needs k, walk hyperparams

Method	Strategy	Complexity	Strength	Limitation
GNN-based	Neural + graph structure	Varies	Soft membership, features	Requires training data or objective design

Takeaway

Divisive and agglomerative heuristics approximate the NP-hard modularity optimum in polynomial time. Girvan–Newman is the didactic divisive method; Louvain and Leiden are the scalable agglomerative methods used in practice. Graph partitioning (min-cut, spectral) targets fixed-k balanced splits; embedding-based methods (node2vec, GNNs) detect communities from representation space and can handle overlapping memberships and node features.

Quick Check

You run Girvan–Newman on a small graph and get a dendrogram with a maximum Q at the cut that produces 4 communities. You then run Louvain on the same graph and it reports 3 communities with a slightly lower Q.

1. Which answer is "correct"?
2. What could explain the discrepancy?

Quick Check — Solution

1. Neither is absolutely correct — both are heuristics approximating the NP-hard modularity optimum. The higher Q value is weak evidence, not proof, and the difference may be within noise.

2. Possible explanations:

- Louvain is greedy and can get stuck in local optima that miss small communities.
- Girvan–Newman explores more cuts but is sensitive to the order of edge removals.
- Modularity itself has a resolution limit; both methods share this blind spot.
- The graph may genuinely admit multiple near-optimal partitions — real structure can be ambiguous.

The honest practice is to run multiple algorithms, report agreement, and treat disagreement as a signal about the graph itself.

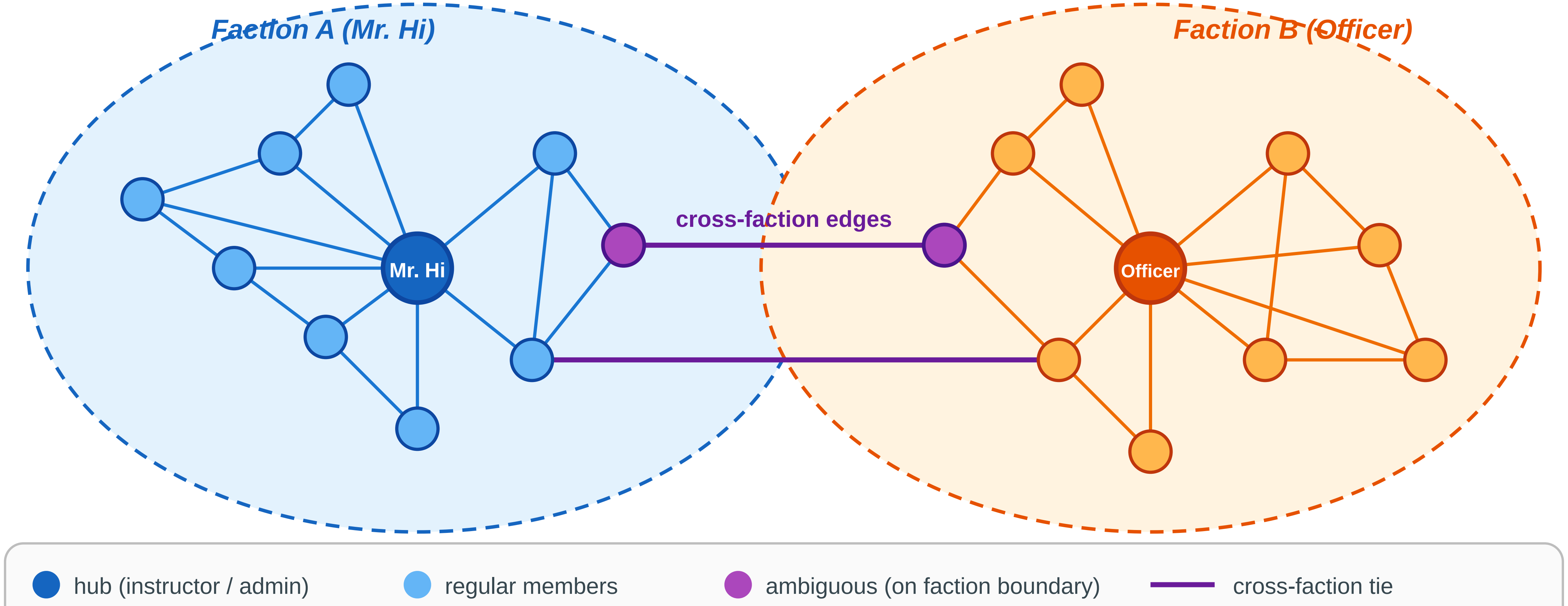
Empirical Anchor — Zachary's Karate Club

Guiding Question: Do community-detection heuristics actually recover real social groups — and how do we know?

The Story

Zachary's Karate Club — the community-detection benchmark

34 members · pre-split friendships · two factions formed around Mr. Hi and the Officer



Schematic — the real Zachary graph has 34 nodes and 78 edges; the split is the documented outcome.

Zachary (1977), *An Information Flow Model for Conflict and Fission in Small Groups*, [J. Anthropological Research 33, 452–473](#).

Wayne Zachary observed a university karate club for two years. A conflict arose between the instructor (**Mr. Hi**) and the chief administrator (**Officer**), and the club eventually **split in two** — with each member following one or the other.

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What the Algorithms Find

Method	Typical result
Girvan–Newman	Recovers the two factions with **1 misclassified** node
Louvain / Leiden	Recovers the two factions; typically finds **4 sub-groups** at max Q
Max-modularity (exact on this small graph)	Finds the same 4 sub-groups nested inside the 2 factions

The ~1-node misclassification is structurally meaningful: the node in question genuinely sat on the boundary between the two factions during the conflict period, and which side they joined was a close call even to Zachary himself.

What the Study Could Not See

Even in this iconic success case, the empirical validation has blind spots that parallel Lecture 03's gap between construct and measurement:

1. Modularity resolution limit (Fortunato & Barthélemy 2007). On the 34-node karate graph the exact max-Q partition has **4** communities, not **2**. The ground-truth split is not even the modularity optimum — the algorithm is answering "which partition maximizes Q", not "which partition matches the real factions".

2. Binary ground truth. Zachary's data records which faction each member *eventually* joined — but not the *strength* of their preference, their uncertainty during the conflict, or whether a third "neutral" group briefly existed. The true social structure may have been continuous or overlapping.

3. Static snapshot. The friendships were recorded at a single time. The dynamics of the conflict — who influenced whom, when someone shifted — are invisible to the algorithm.

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Takeaway

> Zachary's karate club shows that community-detection heuristics recover real social structure from graph topology alone — but also that even the "canonical" answer has blind spots: modularity's resolution limit, binary ground truth, and static snapshots all hide real structure the algorithm cannot see.

Wrap-Up — Closing the Modeling Loop

Across Lectures 03 and 04 we have traversed the full modeling/measurement loop twice:

Level	Theoretical ideal	NP-hard recovery	Polynomial proxy	Empirical test
Edges (L03)	Strong/weak labeling	MaxSTC	Clustering coeff., overlap	Kossinets–Watts email
Nodes / Groups (L04)	Modularity-optimal partition	Modularity maximization	Girvan–Newman, Louvain, Leiden	Zachary karate club

Every concept in the two lectures occupies one of those four cells.

Reading

Chapter 3.6 and Chapter 9 of Easley & Kleinberg (2010) cover community structure, brokerage, and centrality. Newman (2010), Chapter 7, gives the formal treatment of centrality measures. For modularity and Louvain, see Blondel et al. (2008) and Traag et al. (2019) for Leiden.

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Image Credits & References

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning About a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/